CS-302: Artificial Intelligence

LECTURE NO 12:

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Organization of Lecture

1. Knowledge Types in A.I
2. Semantic Networks in A.I
3. Frames for Knowledge Representation
4. Scripts for Knowledge Representation
5. Bayes' Theorem
6. Bayesian Belief Network in A.I

1. Knowledge Types in A.I (1/3)

① **Procedural Knowledge:** Describes how to solve Problem.

- ↳ Also known as Imperative Knowledge.
- ↳ Provides direction on how to do something.
- ↳ Can be directly applied to a task.
- ↳ Includes
  - → Rules
  - → Strategies
  - → Procedures
  - → ⋮

② **Declarative Knowledge:** Describes what is known about problem.

- ↳ Tells us facts - what things are.
- ↳ Includes
  - → Concepts
  - → Facts
  - → objects
- ↳ Also called descriptive Knowledge.

1. Knowledge Types in A.I (2/3)

3) **Meta Knowledge:** Describes knowledge about another knowledge.

↳ used to pick other knowledge that is best suited for solving a probelm.

↳ knowledge about types of knowledge.

4) **Heuristic Knowledge.:** It is representing knowledge of some experts in a field or subject.

↳ describes rules of thumb that guides reasoning process.

↳ Previous experiences, approaches.....

1. Knowledge Types in A.I (3/3)

5) **Structural Knowledge:** It is the basic knowledge to probelm solving.

↳ describes relationships b/w various concepts/objects

- kind of
- part of
- group of

↳ describes an expert overall mental model of a probelm.

2. Semantic Networks in A.I (1/3)

- Graphical notation for representing knowledge in interconnected nodes pattern.
- Popular in A.I and Natural Language Processing (NLP) because it represents knowledge or support reasoning.
- Alternative of Predicate Logic.
- Nodes represents objects.
  - Circle (Physical)
  - Ellipse (Concept)
  - Rectangle (Situation)

2. Semantic Networks in A.I (2/3)

↳ Arcs represents Relation b/w Objects.] Links/arrows

↳ Link Labels] specify relationships

↳ Also known as Associative Nets] Show association between Nodes.

Components:

① Lexical Components: Nodes, Links & Labels.

② Structural: Also Links or Nodes] Directed

2. Semantic Networks in A.I (3/3)

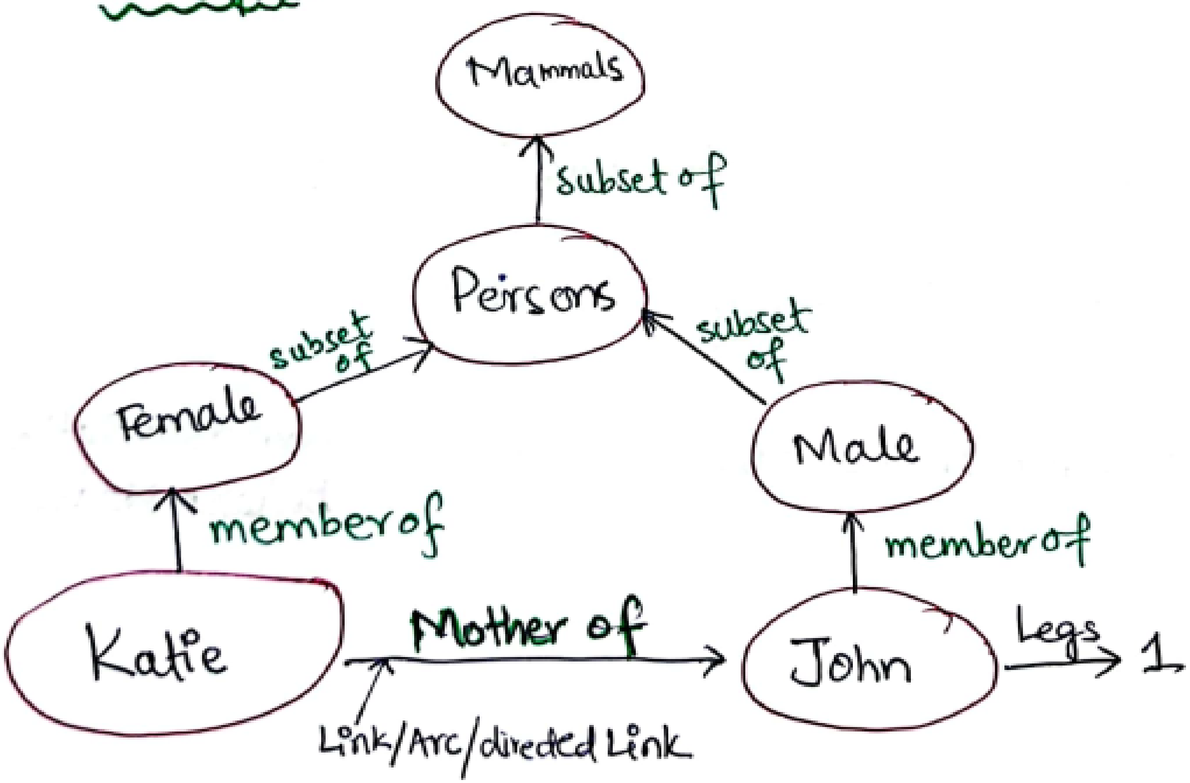
③ Semantic: Definition related to links/nodes. It basically represents the facts.

④ Procedural:

i- Constructor : Creation of new links.

ii- Destructor : Removal of links.

Example:



3. Frames for Knowledge Representation (1/5)

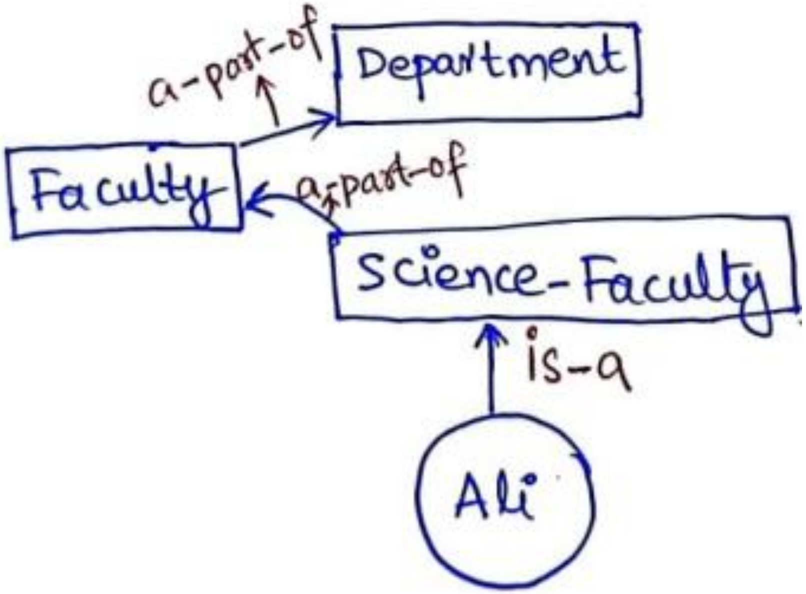
<u>Representing Knowledge using FRAMES:</u> Frame is a collection of attributes or slots and associated values that describe some real-world entity.

↳ uses datastructure (records) to represent the knowledge represented in Semantic N/W.

↳ Each frame represents the node in semantic Network as a class, or an instance and each relation as slot i.e., In semantic N/W:

- Each Node is represented by frame
- Each relation is represented by slot.

3. Frames for Knowledge Representation (2/5)



Attributes attached with each slot: -

① Instance: Relates slot with class.

② Definition: Slot definition or Slot Value.

③ Default: Slot default value.

3. Frames for Knowledge Representation (3/5)

elements

(4) Domain : Slot elements domain . i.e., they belong to integer, characters, alphanumeric etc.

(5) Range: Specifies class of which elements.

(6) Range Constrain: Logical expression that must be True. [$6<i<10$]

(7) To-Compute: Value of slot is to be computed.

(8) Single-Valued: function returns single value.

(9) Inverse: Inverse of slot which is used in reasoning.

(10) Transfers through: It is used in inheritance.

3. Frames for Knowledge Representation (4/5)

Reasoning actions that can be performed using frames:

1. Relating the definition: A definition can be propogated through is a, inverse links to relate all information.

2. Inheritance: Inherited all values including default values of the slot.

3. Legality of Values: It checks the legality of slot value. We can check it using Range constraint.

3. Frames for Knowledge Representation (5/5)

④ <u>Consistency Check</u>: Verifying slot value consistency before adding to the frame. This could be done using Domain constraint.

⑤ <u>Maintaining Consistency</u>: when one slot is updated, its inverse should also be updated.

⑥ <u>Computation of a Value of Slot</u>: If we want to compute the value of any slot, it can be done using:

- to-compute
- transfers through. etc.

4. Scripts for Knowledge Representation (1/4)

SCRIPTS: Used for knowledge Representation.

↳ Script is a structure that prescribes a set of circumstances which could be expected to follow on from one another.

↳ Considered to consist of a no. of slots or frames but with more specialized roles.

Components of a Script:

① Roles: Persons involved in Event. For e.g: a student is going for an exam to examination Hall, here student is a Person, therefore student is an example of Role

② Props: The objects which are involved in event. Same example as in Roles. The objects which student need are pen, university card, answersheet etc.

4. Scripts for Knowledge Representation (2/4)

(3) Entry Conditions: Conditions that need to be satisfied before event occurs in script. For e.g: university card & paperslip must be present with the student.

(4) Results: Conditions that will be true after event in script occurs. For e.g: Solved/Filled answer sheet by student after the question paper was given to student.

(5) Track: Variations on the Script. Different papers exam in same Hall but on different days or dates.

4. Scripts for Knowledge Representation (3/4)

(6) Scenes : Sequence of event that occurs. For e.g:

Exam hall entry

↓

Getting allocated seat

↓

Getting answer sheet

↓

Getting question paper

↓

writing answers

↓

Submit answer sheet

4. Scripts for Knowledge Representation (4/4)

Advantages of Scripts:

↳ ① Event prediction is possible using scripts.

② Single coherent interpretation may be build up of collection of observations, i.e:

Interpretation = Observation 1 + Observation 2 etc.

Disadvantages of Scripts:

↳ ① Less General than frames.

② May not be suitable to represent all kind of knowledge

5. Bayes' Theorem (1/5)

5) **Baye's Theorem:** Describes the probability of an event, based on prior knowledge of conditions that might be related to the event.

↳ In Probability theory it relates the conditional probability & marginal probabilities of two random events.

↳ Calculate $P(B|A)$ with knowledge of $P(A|B)$

As we know that:

$$P(A \cap B) = P(A|B) \cdot P(B) \quad \text{— (i)}$$

$$P(A \cap B) = P(B|A) \cdot P(A) \quad \text{— (ii)}$$

Conditional Probability

Hypothesis, Evidence

$$P(\overset{\text{Hypothesis}}{H}|\overset{\text{Evidence}}{E}) = \frac{\text{no. of time H and E}}{\text{no. of times E}}$$

$$P(H|E) = \frac{P(H \overset{\text{and}}{\cap} E)}{P(E)} \Bigg\} \text{Prob. of H When E is true.}$$

5. Bayes' Theorem (2/5)

From (i) & (ii) L.H.Ss are equal.

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

So, $\boxed{P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}}$ → Baye's Theorem formula

where $P(A|B)$ = Posterior Probability = Probability of A when B is true.

$P(B|A)$ = Likely hood (Probability of B when A is true)

$P(A)$ = Prior Probability (Probability of hypothesis)

$P(B)$ = Marginal Probability (Probability of Evidence).

5. Bayes' Theorem (3/5)

Example: What is the probability that person has disease dengue with neck pain? Data is given below:

* 80% of time dengue causes neck pain
* $P(\text{dengue}) = \frac{1}{30,000}$
* $P(\text{neckpain}) = 0.02$

Solution:

Suppose a = Proposition that Person has neck pain.
& b = " " " " dengue

Given: 80% of time dengue causes neckpain = $P(a|b)$

$$P(a|b) = 80\% = 0.8$$

$$P(\text{dengue}) = P(b) = \frac{1}{30,000}$$

$$P(\text{neckpain}) = P(a) = 0.02$$

5. Bayes' Theorem (4/5)

$P(b|a) = ?$

According to Baye's Theorem:

$$P(b|a) = \frac{P(a|b) \cdot P(b)}{P(a)} = \frac{(0.8) \cdot (33.33 \times 10^{-6})}{0.02} = 1.33 \times 10^{-3}$$

$$P(b|a) = 1.33 \times 10^{-3} = 0.00133$$

$$\boxed{P(b|a) = 0.00133}$$

5. Bayes' Theorem (5/5)

Applications of Baye's Theorem in A.I

1. Robot/ Automatic Machine: next step would be computed when previous step has been completed. "OR" Next step is calculated based on previous step.

2. Forecasting: Weather forecasting

3. Monty Hall Probelm: Monty Hall Probelm can be solved.

6. Bayesian Belief Network in A.I (1/4)

It defines probabilistic independencies and dependencies among the variables in the Network.

↳ It is a probabilistic graphical model which represents a set of variables and their conditional dependencies using a directed acyclic graph.

D A G

DAG

↳ Built from Probability distribution.

↳ Consists of

1. DAG (Directed Acyclic Graph).
2. Table of Conditional Probabilities

6. Bayesian Belief Network in A.I (2/4)

↳ <u>Node:</u> Corresponds to a Random Variable. This Randomnumber can be Continuous or discrete

↳ <u>Arc/Directed arrows:</u> It represents casual relationship or conditional probabilios among random variable.

6. Bayesian Belief Network in A.I (3/4)

Example 1

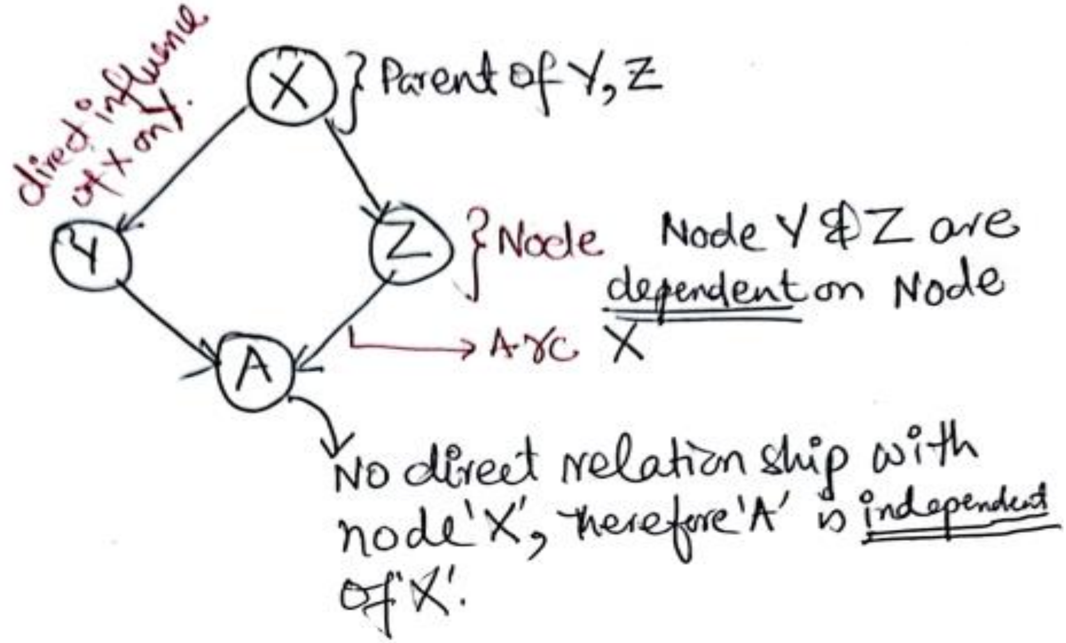


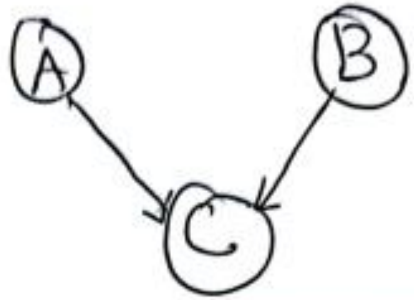
Table:-

→ the table is made. which contains conditional probabilities of all the nodes with effect of their Parent nodes.

→ To propagate belief in Bayesian N/w, initial "Directed Acyclic Graph" is converted into an undirected graph in which the arcs can be used to transmit probabilities in direction of Evidence.

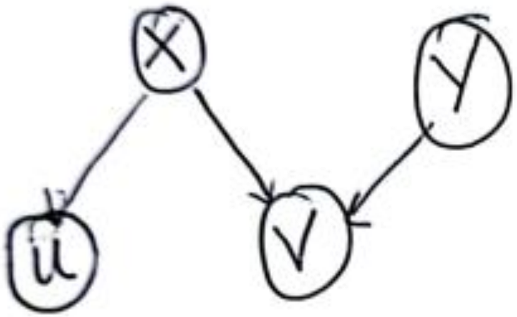
6. Bayesian Belief Network in A.I (4/4)

T	0.002
F	0.998



T	0.001
F	0.999

A	B	P(C=T)	P(C=F)
T	T	–	–
T	F	–	–
F	T	–	–
F	F	–	–



↳ there should not be a cycle

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7. *Related documents from open source, mainly internet.*